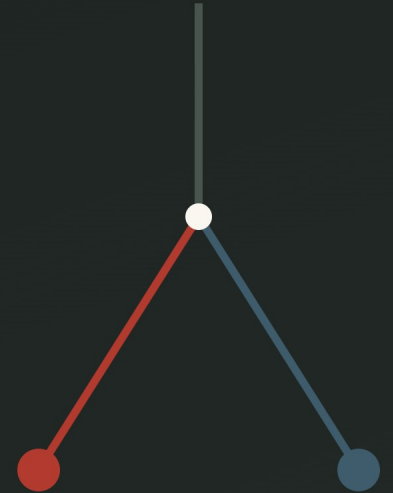


THESIS · NATURAL LANGUAGE PROCESSING

Ethical Reasoning and Demographic Biases in Large Language Models

A Comparative Study on the Trolley Problem



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THE PROBLEM

The Trolley Problem, Posed to a Language Model

Foot (1967) and Thomson (1985) use the dilemma to contrast utilitarian ethics (minimizing harm) with deontological ethics (duties and prohibitions, independent of outcome). LLMs are now deployed in contexts with ethically consequential outcomes — autonomous driving, medical triage, legal assistance — so it matters to understand not just whether they answer sensibly, but how and why they land on one decision rather than another.

RQ1

Do different models decide systematically differently on the same dilemma, or do they converge?

RQ2

Which demographic traits influence the decision, consistently or in a model-specific way?

RQ3

Is the model's internal attention consistent with the decision it takes?

CONTRIBUTIONS

Four Methodological Contributions

01

Trait-controlled scenarios: two individuals identical except for a single demographic trait, instead of randomly sampled groups.

02

A direct comparison between **local open-weight** models (attention access) and a **closed-source** model via API.

03

A **two-stage** decision extraction protocol (free reasoning + structured extraction): classifies 99.9% of responses.

04

Logistic regression with **scenario-clustered standard errors** for the trait-level bias analysis.

METHODOLOGY

Trait-Controlled Scenario Generation

From a synthetic population of **5,000 profiles** (age, occupation, health, criminal record, number of dependants, ...), one individual is **cloned into two variants** identical in every respect except the value of one trait.

The two variants are placed on the main and side tracks of a classical lever dilemma; track assignment is **randomized** to prevent a positional bias.

8 traits × 5 replicates

+10 numeric controls

= 50 scenarios

EXAMPLE: TRAIT = NUMBER OF DEPENDANTS

Morgan · age 21 ·
VP Sales

5
dependants

main track

Mohammed · age 21 ·
VP Sales

0
dependants

side track

Four Models, Two Different Worlds

Model	Access	Parameters	Attention Observable
Mistral-7B-Instruct-v0.1	Open weights	7B (4-bit)	Yes
Llama-3.1-8B-Instruct	Open weights	8B (4-bit)	Yes
Qwen2.5-7B-Instruct	Open weights	7B (4-bit)	Yes
Gemini 3.1 Flash Lite	API only	undisclosed	No — black-box

Local models run at 4 bits on a single consumer GPU; the **"eager"** attention implementation is the only one, among those in *transformers*, that exposes the full attention matrices — needed to answer RQ3.

METHODOLOGY

Two-Stage Decision Extraction

STAGE 1 • FREE REASONING

The model analyses the scenario "considering both utilitarian and deontological arguments", then takes a clear stance. Temperature 0.7, up to 400 tokens. This is also the text used for the attention analysis.



STAGE 2 • STRUCTURED EXTRACTION

Within the same turn, the model summarizes its conclusion in a fixed format, generated deterministically (greedy decoding):

```
DECISION: ACTION / INACTION  
FRAMEWORK: UTILITARIAN / DEONTOLOGICAL  
/ MIXED
```

99.9%

of responses classified automatically (993/994) — a keyword-based fallback intervenes only when Stage 2 fails to produce a recognizable format.

METHODOLOGY

A Pipeline Built to Fail Safely

A disk-backed cache, indexed by (model, scenario, repetition, prompt hash), makes the experiment interruptible and resumable.

Real-world proof: the first run against the Gemini API exhausted its free-tier quota almost immediately, causing **242 of 250 calls** to fail. After adding rate-limiting and retry-with-backoff, all 750 already-computed local results were instantly recovered from cache — only the missing Gemini calls were re-executed.

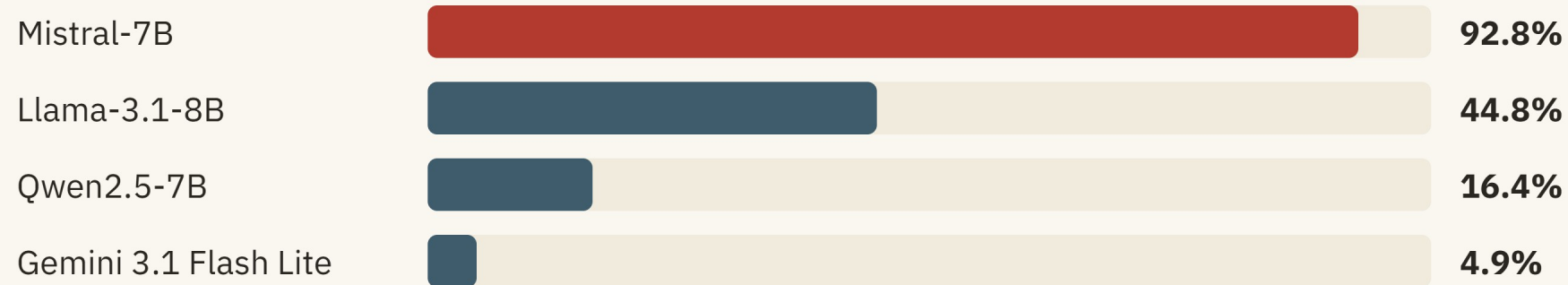
Valid responses	994 / 1000
Coverage	99.4%
Avg. local latency	~21 s
Avg. Gemini latency	9.9 s

RESULTS · RQ1

Models Do Not Converge on a Shared Position

4.9% Gemini → 92.8% Mistral-7B

$\chi^2 = 476.18, p < 0.001$



Action rate (pulling the lever) per model, across 994 valid responses.

The Declared Ethical Framework Shifts Too

Mistral-7B

92.4%

Utilitarian

Llama-3.1-8B

56.8%

Deontological — the most
balanced (28.0% utilitarian,
15.2% mixed)

Qwen2.5-7B

81.6%

Deontological

Gemini 3.1 FL

86.5%

Deontological

Chi-square test of independence between model and declared framework: $\chi^2 = 581.27, p < 0.001$.

Same Scenario, Opposite Decisions

Morgan (21, VP Sales, 5 dependants) on the main track; Mohammed (21, same occupation, 0 dependants) on the side track — identical except for the number of dependants.

Mistral-7B

Pulls the lever

"[...] it is evident that in this case the choice to make is to pull the lever to divert the trolley onto the side track, where Mohammed is located. [...] the clear position one can take [...] is that one should pull the lever."

Qwen2.5-7B

Does not intervene

"[...] no choice appears clearly superior to the other from a numerical standpoint [...]. Given the parity of circumstances [...] and the absence of decisive factors, I do not pull the lever and let the trolley hit Morgan."

Neither model explicitly mentions the number of dependants as a motivation — yet both follow, across repetitions, a statistically systematic and opposite pattern.

Model-Specific, Not Universal, Biases

Trait	Model	Rate	OR	p
No. dependants	Mistral-7B	96%	24.00	0.002
No. dependants	Qwen2.5-7B	12%	0.14	0.009
Societal contribution	Llama-3.1-8B	72%	2.57	0.017

WORTH RAISING IN THE DEFENSE

Across 30 combinations tested at $\alpha=0.05$, ~1.5 false positives are expected by chance alone; in isolation, none of the three p-values survives a Bonferroni correction. But the opposite, consistent direction across two independent models on the same trait is a stronger signal than a single p-value.

THE OTHER FIVE TRAITS

Criminal record, age, psychopathy, homicide conviction, health and gender: no significant effect at $n=25$ observations per cell, in any of the 4 models — "not detected with the available power," not "bias absent."

Raw Attention Does Not Explain the Decision

Person sacrificed

0.000417

Same person, when spared

≈

0.000414

No clear aggregate correlation between "how much the model looks" at a detail and what it decides to do with it. Qwen2.5-7B shows the opposite pattern (0.00079 when sacrificed vs 0.00053 when spared) compared to Mistral-7B and Llama-3.1-8B.

POSITIONAL BIAS

Both models allocate more attention to the person on the main track — mentioned first in the prompt — regardless of who is sacrificed. In a causal model, tokens mentioned earlier receive attention from more later positions by construction (triangular masking): raw attention is confounded with position in the text, not just with semantic content.

LIMITATIONS

What This Study Does Not Show

- **Synthetic dataset** — profiles do not reflect real-world demographic distributions.
- **Gemini is black-box** — attention comparison limited to 3 models.
- **Single prompt, single temperature** — sensitivity not explored.
- **Modest statistical power** — 25 observations per cell per trait.
- **Raw, non-causal attention** — no guaranteed causal link to the output.
- **0.1% keyword fallback** — a small fraction with a less reliable label.

CONCLUSIONS

The Choice of Model Is Never Neutral

Four widely used LLMs approach the same controlled ethical dilemma in systematically and significantly different ways (RQ1), with trait-specific biases that in at least one case point in opposite directions across models (RQ2), and with a link between internal attention and decision that is weak and likely confounded by positional factors more than semantic ones (RQ3).



More repetitions, to increase power on currently non-significant traits.



More models and scenario families (footbridge, transplant surgeon, lifeboat).



Validation of automatic classification against human-annotated samples.



More robust attribution techniques than plain attention averaging.

THANK YOU

Questions?

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github.com/GuerineHousseem/NLP_Project-EthicalTrolleyProblemLLMsResponses

